Test Case Document (Updated)

Smart Complaint Management System · Team 5D
Version 2.0 - after Sprint 2

Version	1.0
Date of execution	12 Nov 2026, 15:45
Commit tested	66d1420
Branch	main
Test framework	pytest, pytest-cov (coverage.py), pytest-html
Overall result	114 of 114 automated tests passed

1. What changed

This updates the Test Plan test cases (v1.0, TC01-TC20) to match what was built in Sprints 1 and 2. Expected results were revised where the design changed, TC21-TC28 were added for new behaviour, and every case now records its actual result, status and the automated tests that execute it. Status is computed from the latest automated run: a case passes only if all of its tests pass. Concurrency cases UT-04 to UT-09 / IT-04 for Sprint 3 Task 1 are kept in Documents/Sprint_3_Test_Cases.md on branch feature/concurrency-safe-storage.

Summary: 23 Pass, 0 Fail, 1 Partial, 3 Not Implemented, 1 Not Executed (of 28).

2. Test cases

ID	Req	Scenario	Preconditions	Test steps	Test data	Expected result	Actual result	Pri.	Status
TC01	FR-001	Submit complaint with valid details	Citizen is logged in and on the home screen.	1. Open New Complaint. 2. Enter title and description. 3. Select category and severity. 4. Attach an image and enable GPS. 5. Submit.	Title: Pothole on MG Road; category: Roads; severity: High; image: 2 MB PNG; GPS 12.9716, 77.5946.	Complaint saved with all fields; unique SCMS-YYYYMMDD-NNNNN ticket ID shown on confirmation.	Complaint stored with title, description, severity, category, image and GPS; ticket ID in SCMS-YYYYMMDD-NNNNN format returned and shown.	High	Pass
TC02	FR-001	Validate mandatory complaint fields	Citizen is logged in; complaint form open.	1. Leave title, description and severity empty. 2. Submit. 3. Review validation feedback.	Title: empty; description: empty; severity: none.	Submission blocked; each missing field reported with a message; nothing stored.	HTTP 400 with field errors for title, description and severity; no complaint stored.	High	Pass
TC03	FR-001	Reject oversized or invalid complaint image	Citizen is logged in; complaint form open.	1. Attach an image over 5 MB. 2. Submit. 3. Repeat with a non-image file and with 6 images.	Image: 5 MB + 8 bytes PNG; a renamed executable; 6 images.	Image rejected with a message naming the 5 MB limit; non-images and more than 5 images rejected.	Oversized image rejected ('exceeds the 5 MB limit'); non-image and 6-image uploads rejected.	High	Pass
TC04	FR-001	Enforce title (120) and description (2000) limits	Citizen is logged in; complaint form open.	1. Submit with exactly 120 / 2000 characters. 2. Submit with 121-character title. 3. Submit with 2001-character description.	Title: 120 and 121 chars; description: 2000 and 2001 chars.	Boundary values accepted; values beyond the limits rejected.	120/2000 accepted (201); 121-char title and 2001-char description rejected (400).	High	Pass

ID	Req	Scenario	Preconditions	Test steps	Test data	Expected result	Actual result	Pri.	Status
TC05	FR-002	Suggest top-3 categories with confidence	NLP model trained and loaded.	1. Enter categorisable text. 2. Request suggestions. 3. Review top 3 and confidence. 4. Accept or override.	Text: water leakage from main pipe on 5th cross.	Top-3 categories with confidence scores, correct category first; citizen can accept or override.	Three suggestions in descending confidence; 'Water Supply' ranked first; override saved.	High	Pass
TC06	FR-002	Handle low-confidence (< 70%) NLP suggestions	NLP model loaded; complaint form open.	1. Enter text for which every suggestion is below 70%. 2. Review the response. 3. Choose a category manually.	Ambiguous text; all confidences < 70%.	System asks the citizen to choose a category manually and the submission continues.	No low-confidence prompt; the citizen can still pick any category manually.	Medium	Not Implemented
TC07	FR-003	Generate unique ticket ID after submission	Citizen is logged in.	1. Submit a valid complaint. 2. Note the ticket ID. 3. Submit again and compare. 4. Repeat on a new day.	Many sequential submissions; a submission on the next day.	Each submission gets a unique SCMS-YYYYMMDD-NNNNN ID; sequence restarts each day.	Sequential IDs unique and correctly formatted; sequence restarts on a new day.	High	Pass
TC08	FR-004	Route complaint to department and agent	Agents exist in several departments and zones.	1. Submit with a known category and zone. 2. Let routing run. 3. Inspect department and agent.	Category: Water Supply; zone: Zone 3; agents in Zone 1 and Zone 3 with different loads.	Correct department; same-zone agent preferred; least-loaded agent chosen.	Routed to Water Department, Zone 3 agent; least-loaded agent chosen on ties.	High	Pass
TC09	FR-004	Route complaint when no agent is available	No active agent in the target department; a supervisor exists.	1. Submit for that department. 2. Inspect assignment. 3. Check supervisor notifications.	Category: Electricity; no Electricity agent.	Complaint stays New, flagged for the supervisor queue; supervisors alerted.	Unassigned, needs_supervisor = true, status New; supervisor received 'needs_supervisor' alert.	High	Pass
TC10	FR-005	Lifecycle with timestamped history	Citizen and assigned agent accounts.	1. Submit. 2. Move through In Progress, Pending Info, In Progress, Resolved. 3. Citizen closes. 4. Open history. 5. Try an invalid jump.	One complaint; every transition.	New -> Assigned -> In Progress -> Pending Info -> Resolved -> Closed, each logged with time and actor; invalid transitions rejected.	All 7 history entries present in order with timestamps and actor; invalid jump rejected (400).	High	Pass
TC11	FR-006	SLA warning when deadline is near	Complaint with an SLA deadline; assigned agent.	1. Create a High complaint (24 h SLA). 2. Set remaining time to 90 min. 3. Run the SLA check.	SLA 24 h; 90 minutes remaining.	One warning alert to the assigned agent when < 2 h remain.	Single warning alert raised and delivered to the agent's inbox.	High	Pass
TC12	FR-006	Alert when SLA is breached	Open complaint past its deadline.	1. Move the deadline into the past. 2. Run the SLA check twice. 3. Check supervisor alerts and dashboard.	Deadline 5 minutes ago.	One breach alert to supervisors; no duplicate on re-check; counted as overdue.	Breach alert raised once; dashboard overdue count = 1.	High	Pass
TC13	FR-007	Notifications on status changes	Citizen with email on; SMS opt-in with phone.	1. Change status. 2. Check in-app inbox, email and SMS. 3. Resolve and check resolution message.	Status changes and resolution of one complaint.	In-app + email (and SMS if opted in) sent for each change, within 60 s.	In-app and email sent on each change; SMS only for opted-in users; provider failure does not break the request.	Medium	Pass
TC14	FR-008	Save resolution rating and feedback	Complaint resolved.	1. Open feedback prompt. 2. Give 1-5 stars and a comment. 3. Submit.	Rating 4; comment 'Fixed well'.	Rating saved; CSAT updated.	No rating endpoint or prompt exists.	Medium	Not Implemented
TC15	FR-008	Auto-close unrated complaint after 7 days	Resolved, unrated complaint.	1. Resolve. 2. Wait 7 days (simulated). 3. Check status.	Resolved 7 days ago, no rating.	Complaint auto-closed.	Complaint remains Resolved.	Medium	Not Implemented

ID	Req	Scenario	Preconditions	Test steps	Test data	Expected result	Actual result	Pri.	Status
TC16	FR-009	Escalate a stalled complaint	Supervisor logged in; open complaint.	1. Open stalled list. 2. Escalate with reason and receiving team. 3. Check queue, SLA, notifications.	Reason: No response from field team; team: Water Department.	Moved to priority queue with a reset SLA; citizen and agents notified.	Ticket in priority queue, new deadline later than old, notifications sent.	High	Pass
TC17	FR-010	Role-based access control	Citizen, Agent, Supervisor and Admin accounts.	1. Log in as each role. 2. Call each protected endpoint. 3. Call without a token.	9 protected endpoints x 4 roles + anonymous.	Allowed roles succeed; other roles get 403; no token gets 401.	All 45 role/endpoint combinations returned the expected 2xx / 403 / 401.	High	Pass
TC18	FR-011	Dashboard metrics accuracy and refresh	Complaints with known states.	1. Create a known dataset. 2. Open dashboard. 3. Compare counts, SLA compliance. 4. Check refresh.	2 complaints, 1 resolved on time.	Metrics match stored data; page refreshes every 60 s.	total 2, open 1, resolved 1, SLA compliance 100%; 60 s auto-refresh present.	Medium	Pass
TC19	FR-012	Immutable audit log of complaint actions	Complaint with actions by several roles.	1. Perform actions. 2. Open the audit log. 3. Check actor and time. 4. Try to alter an entry.	Status change, note, escalation.	Every action logged with actor ID and time; log immutable.	Actor + timestamp recorded per action inside the complaint; no immutability guarantee.	High	Partial
TC20	NFR-001	500 concurrent users; API p99 < 500 ms	Deployed test environment and load-testing tool.	1. Simulate 500 concurrent users. 2. Measure API p99 and page-load p95.	500 virtual users.	API p99 < 500 ms; page load p95 < 2 s.	Smoke only: see SV-10 in the System Validation Report.	High	Not Executed
TC21	FR-014	Agent uploads evidence files	Agent assigned to an open complaint.	1. Upload a PNG and a PDF. 2. Upload a file over 10 MB. 3. Upload as a non-assigned agent. 4. Upload to a closed complaint.	after.png, report.pdf; 10 MB + 9 bytes PDF.	PNG/JPEG/PDF up to 10 MB saved and linked; oversize, wrong type, wrong agent and closed complaint rejected.	Valid files stored; oversize 400; other agent 403; closed complaint rejected.	Medium	Pass
TC22	FR-013, FR-014	Internal notes and resolution notes	Agent assigned; citizen owns the complaint.	1. Add internal note. 2. View as citizen. 3. Resolve without notes. 4. Resolve with notes.	Note: Valve checked; resolution: Fixed.	Note saved with author and time, hidden from citizen; resolving needs notes; citizen notified.	Note stored with author Ravi and time; hidden from citizen; resolve without notes 400, with notes 200.	High	Pass
TC23	FR-005	Citizen closes or reopens a resolved complaint	Complaint Resolved.	1. Citizen reopens. 2. Check resolution cleared. 3. Resolve again and close.	One resolved complaint.	Reopen returns it to In Progress; close moves it to Closed.	Reopen and close both work; old resolution cleared on reopen.	Medium	Pass
TC24	FR-011	Dashboard filters and CSV report	Supervisor logged in; complaints exist.	1. Filter by date, department, agent. 2. Download CSV report.	Mixed complaints.	Filtered KPIs correct; CSV has a header and one row per complaint.	Filters applied correctly; CSV downloaded with all rows.	Low	Pass
TC25	FR-007	Notification preferences	Logged-in user.	1. Turn email off. 2. Turn SMS on without a phone. 3. Add an invalid phone. 4. Add a valid phone.	Phone: 12345 and +919876543210.	Email opt-out respected; SMS requires a valid E.164 phone.	Invalid inputs rejected with messages; preferences applied to later notifications.	Low	Pass
TC26	FR-010, NFR-002	Authentication security	Registered users.	1. Wrong password. 2. Forged JWT. 3. Expired JWT. 4. Self-register as admin. 5. Read another citizen's complaint. 6. Upload an executable renamed .png.	As listed.	All attempts rejected; passwords stored as bcrypt hashes; role forced to Citizen.	401 for wrong password, forged and expired tokens; role forced to citizen; 403 on other's complaint; executable rejected.	High	Pass
TC27	FR-006	Admin configures SLA rules	Admin and non-admin users.	1. Admin updates High to 12 h. 2. Non-admin tries. 3. Submit a High complaint.	High: 12 h.	Only admin can change rules; new complaints use them.	Admin change applied to new complaints; others get 403.	Medium	Pass

ID	Req	Scenario	Preconditions	Test steps	Test data	Expected result	Actual result	Pri.	Status
TC28	FR-002, FR-004	Category override re-routes the complaint	Complaint New or Assigned.	1. Change the category. 2. Inspect department and agent. 3. Check new agent's inbox.	Roads -> Water Supply.	Re-routed to the new department; new agent notified.	Department changed to Water Department; new agent notified.	Medium	Pass

3. Changes since Test Plan v1.0 and automation mapping

ID	Change	Automated tests (tests/...)	Status
TC01	Confirmation shows ticket ID and category; the timeline is on the Track page (/track) rather than a link on the confirmation.	test_submission::test_complaint_submitted_successfully test_submission::test_images_uploaded_correctly test_submission::test_gps_coordinates_stored test_system_validation::test_sv01_complaint_end_to_end_happy_path	Pass
TC02	Category is no longer mandatory: if the citizen leaves it empty, the top NLP suggestion is used (FR-002). Expected result updated accordingly.	test_submission::test_validation_errors_shown test_system_validation::test_sv02_submission_validation_and_limits	Pass
TC03	Added: file type is checked from file content (JPEG/PNG signature), not the extension; max 5 images.	test_submission::test_oversized_image_rejected test_submission::test_non_image_and_too_many_images_rejected test_system_validation::test_sv02_submission_validation_and_limits	Pass
TC04	Boundary (exact-limit) values added to the test data.	test_system_validation::test_sv02_submission_validation_and_limits	Pass
TC05	None.	test_categorizer::test_top_3_categories_suggested test_categorizer::test_confidence_score_present test_categorizer::test_correct_category_ranked_first test_categorizer::test_user_can_override_suggestion test_system_validation::test_sv03_nlp_suggestions_and_override	Pass
TC06	The 70% threshold is not implemented on main: suggestions are always shown and the top one is pre-selected. Manual selection is possible (verified in SV-03). Planned as Sprint 3 Task 5.	Manual / not automated	Not Implemented
TC07	Sequence restart per day added to the expected result. Concurrent submission is covered by Sprint 3 UT-04 (see defect DEF-01 in the System Validation Report).	test_ticket_id::test_format_is_date_based test_ticket_id::test_sequence_increments_within_a_day test_ticket_id::test_sequence_restarts_on_new_day test_ticket_id::test_no_duplicates_over_many_complaints test_ticket_id::test_submitted_complaint_gets_ticket_id	Pass
TC08	None.	test_routing::test_correct_department_for_each_category test_routing::test_agent_assignment_is_automatic test_routing::test_only_agents_from_the_right_department test_routing::test_workload_balancing_picks_least_loaded test_routing::test_same_zone_agent_preferred test_system_validation::test_sv01_complaint_end_to_end_happy_path	Pass
TC09	None.	test_routing::test_unavailable_agents_go_to_supervisor_queue test_notifications::test_unassigned_complaint_notifies_supervisors test_system_validation::test_sv04_unassignable_complaint_goes_to_supervisor_then_escalated	Pass
TC10	Added: citizen may reopen a Resolved complaint (Resolved -> In Progress); see TC23.	test_status::test_new_complaint_starts_new_then_assigned test_status::test_status_changes_recorded_with_timestamps test_status::test_timeline_shows_complete_history test_status::test_invalid_transition_rejected test_system_validation::test_sv01_complaint_end_to_end_happy_path	Pass
TC11	SLA hours depend on severity only (Critical 4 h, High 24 h, Medium 72 h, Low 120 h); per-category rules are Sprint 3 Task 6.	test_sla::test_deadline_calculated_from_severity_rules test_sla::test_warning_alert_triggers_near_deadline test_system_validation::test_sv05_sla_warning_and_breach_alerts	Pass

ID	Change	Automated tests (tests/...)	Status
TC12	None.	test_sla::test_breach_alert_triggers_after_deadline test_sla::test_resolved_complaints_do_not_alert test_system_validation::test_sv05_sla_warning_and_breach_alerts	Pass
TC13	Notifications are created within the same request (well under 60 s). Email/SMS providers are stubbed in tests; live delivery depends on SMTP/SMS credentials.	test_notifications::test_status_updates_sent_to_citizen test_notifications::test_resolution_notification_sent test_notifications::test_sms_only_for_opted_in_users test_notifications::test_provider_failure_does_not_break_request test_system_validation::test_sv01_complaint_end_to_end_happy_path	Pass
TC14	Rating/feedback is not implemented on main; citizens close or reopen instead. CSAT shows N/A. Planned as Sprint 3 Task 2.	Manual / not automated	Not Implemented
TC15	Not implemented on main. Planned as Sprint 3 Task 2.	Manual / not automated	Not Implemented
TC16	Added: unassigned complaints can be escalated to a team; resolved ones cannot.	test_escalation::test_escalation_moves_ticket_to_priority_queue_with_new_sla test_escalation::test_stakeholders_notified test_escalation::test_escalate_to_another_team test_escalation::test_only_supervisors_and_admins_escalate test_escalation::test_resolved_complaint_cannot_be_escalated test_system_validation::test_sv04_unassignable_complaint_goes_to_supervisor_then_escalated	Pass
TC17	Admin user management covers create and list; update/deactivate is Sprint 3 Task 4.	test_auth::test_role_permissions_enforced test_auth::test_complaint_list_restricted_to_staff test_status::test_only_involved_users_can_view test_system_validation::test_sv06_rbac_matrix	Pass
TC18	CSAT trend shows N/A until ratings exist (TC14).	test_dashboard::test_kpis_are_accurate test_dashboard::test_empty_data_has_no_crashes test_dashboard::test_dashboard_page_has_auto_refresh test_system_validation::test_sv09_dashboard_reflects_stored_data	Pass
TC19	Status history, notes and escalations each record actor and timestamp, but there is no separate tamper-proof log and no audit-log view. Planned as Sprint 3 Task 3.	Manual / not automated	Partial
TC20	Not executed: needs a deployed environment. Single-client in-process smoke test added (SV-10).	Manual / not automated	Not Executed
TC21	New in Sprint 2 (#8).	test_evidence::test_images_and_pdfs_upload_successfully test_evidence::test_file_size_validated test_evidence::test_invalid_files_rejected test_evidence::test_only_assigned_agent_or_supervisor_can_upload test_evidence::test_closed_complaint_rejects_evidence test_system_validation::test_sv08_notes_and_evidence	Pass
TC22	New in Sprint 2 (#7).	test_resolution::test_internal_notes_stored_with_author_and_timestamp test_resolution::test_internal_notes_hidden_from_citizen test_resolution::test_resolving_without_notes_is_rejected test_resolution::test_agent_resolves_with_notes_and_citizen_is_notified test_system_validation::test_sv08_notes_and_evidence	Pass

ID	Change	Automated tests (tests/...)	Status
TC23	New behaviour added in Sprint 2.	test_resolution::test_reopen_clears_resolution_and_done_items_listed_last test_system_validation::test_sv08_notes_and_evidence test_system_validation::test_sv01_complaint_end_to_end_happy_path	Pass
TC24	New in Sprint 2 (#12).	test_dashboard::test_filters test_dashboard::test_report_generated_as_csv test_dashboard::test_dashboard_endpoint_and_permissions test_system_validation::test_sv09_dashboard_reflects_stored_data	Pass
TC25	New in Sprint 2 (#11).	test_notifications::test_email_opt_out_and_preference_validation test_notifications::test_sms_only_for_opted_in_users test_notifications::test_mark_read_and_privacy	Pass
TC26	New: security validation for Test Plan section 5.1.	test_auth::test_password_is_hashed_with_bcrypt test_auth::test_wrong_password_rejected test_auth::test_jwt_authentication_works test_auth::test_self_registration_cannot_pick_admin_role test_system_validation::test_sv07_authentication_security	Pass
TC27	New in Sprint 2 (#9).	test_sla::test_admin_configures_rules_others_cannot test_sla::test_new_complaint_gets_sla_deadline	Pass
TC28	New in Sprint 1/2.	test_notifications::test_reassigned_agent_notified_on_override test_system_validation::test_sv03_nlp_suggestions_and_override	Pass

4. Traceability (requirement -> test cases)

Requirement	Test cases
FR-001 Submission	TC01, TC02, TC03, TC04
FR-002 NLP categories	TC05, TC06, TC28
FR-003 Ticket ID	TC07
FR-004 Routing	TC08, TC09, TC28
FR-005 Lifecycle	TC10, TC23
FR-006 SLA	TC11, TC12, TC27
FR-007 Notifications	TC13, TC25
FR-008 Rating / auto-close	TC14, TC15
FR-009 Escalation	TC16
FR-010 RBAC	TC17, TC26
FR-011 Dashboard	TC18, TC24
FR-012 Audit log	TC19
FR-013 Resolution	TC22
FR-014 Notes / evidence	TC21, TC22

Requirement	Test cases
NFR-001 Performance	TC20
NFR-002 Security	TC26
NFR-003 Availability	-